CS383 · SOFTWARE ENGINEERING

Mock Midterm Solutions

A practice exam in the midterm's format · 60 minutes · 20 marks

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01 Solution Q1: MCQ

#	Answer	Reason
1	**A payroll system built for one specific company**	Customized = built for one specific customer. The others are generic.
2	**Pay as you go**	From the Benefits column of the SaaS slide.
3	**Sprint goal and sprint backlog**	Team capacity and the product backlog are inputs.
4	**Speed**	The factors are consistency, respect, inclusion and honesty.

02 Solution Q2: ethics

SEEPP ethic example	Principle	Reason
Review the work of others in an objective, candid, and properly-documented way.	**COLLEAGUES**	Fair treatment of colleagues' work.
Identify, document, and report significant issues of social concern... to the employer or the client.	**CLIENT AND EMPLOYER**	The reporting is directed to the employer or the client.
Ensure that software engineers know the employer's policies and procedures...	**MANAGEMENT**	The manager is the one who makes sure the team knows the policies.
Not promote their own interest at the expense of the profession, client or employer.	**PROFESSION**	Protecting the integrity and reputation of the profession.

The third statement starts with "Ensure that software engineers...". That is the pattern of Management statements, because it speaks of a manager's duty towards the engineers.

03 Solution Q3: sprint calculations

1) Sprint Velocity $= \frac{20 + 26 + 24 + 22 + 28}{5} = \frac{120}{5} = 24$

2) Focus Factor $= \frac{24}{4 \times 8} = \frac{24}{32} = 0.75$

3) Working days = 2 weeks × 5 days = 10 days

Capacity = 0.75 × (4 × 8 × 10) = 0.75 × 320 = 240 hours

4) 26 > 24 → Prudent

The backlog (26) is slightly larger than the velocity (24), so the plan is prudent: possible with caution, and it is better to bring it down to about 24 points.

04 Solution Q4: process model

Project	Answer	Evidence in the text
1	**Component-Based**	It must integrate a gateway, an API and a ready component.
2	**Plan-Driven**	Safety-critical, fixed requirements, a large team, documentation and sign-offs.
3	**Agile**	Daily feedback, weekly change, 4 developers, a short deadline.
4	**Plan-Driven**	Fixed regulation, requirements set up front, 25 people, a high penalty for defects.

05 Solution Q5: Product / Organization / External

#	Answer	Reason
1	**Product**	The security of the product itself.
2	**External**	A standard imposed by law.
3	**Organization**	The company's operating environment.
4	**Product**	The performance of the product.
5	**Organization**	An internal development procedure of the company.
6	**External**	A requirement from a government body.